12.4.2 PxIN Register

Port X Input Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-2. PxIN Register

7	6	5	4	3	2	1	0
PxIN							
r	r	r	r	r	r	r	r

Table 12-5. PxIN Register Description

Bit	Field	Type	Reset	Description
7-0	PxIN	R	Undefined	Port X input 0b = Input is low 1b = Input is high

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

12.4.5 PxREN Register

Port X Pullup or Pulldown Resistor Enable Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-5. PxREN Register

7	6	5	4	3	2	1	0
PxREN							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-8. PxREN Register Description

Bit	Field	Type	Reset	Description
7-0	PxREN	RW	0h	Port X pullup or pulldown resistor enable. When the port is configured as an input, setting this bit enables or disables the pullup or pulldown. 0b = Pullup or pulldown disabled 1b = Pullup or pulldown enabled

12.4.7 PxSEL0 Register

Port X Function Selection Register 0 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-7. PxSEL0 Register

7	6	5	4	3	2	1	0
PxSEL0							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-10. PxSEL0 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL0	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. See PxSEL1 for the definition of each value.

12.4.8 PxSEL1 Register

Port X Function Selection Register 1 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-8. PxSEL1 Register

7	6	5	4	3	2	1	0
PxSEL1							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-11. PxSEL1 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL1	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. 00b = General-purpose I/O is selected 01b = Primary module function is selected 10b = Secondary module function is selected 11b = Tertiary module function is selected

12.4.9 PxSELC Register

Port X Complement Selection (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-9. PxSELC Register

7	6	5	4	3	2	1	0
PxSELC							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-12. PxSELC Register Description

Bit	Field	Type	Reset	Description
7-0	PxSELC	RW	0h	Port selection complement. Each bit that is set in PxSELC complements the corresponding respective bit of both the PxSEL1 and PxSEL0 registers; that is, for each bit set in PxSELC, the corresponding bits in both P1xEL1 and PxSEL0 are both changed at the same time. Always reads as 0.